

UV Finiteness and Renormalization Group Flow of Curvature-Memory Scalar-Tensor Gravity

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Abstract

This is Paper III of a four-paper series on Curvature-Memory Scalar-Tensor Gravity (CMSTG). We establish the ultraviolet structure of CMSTG at one and two loops in the resummed propagator. The theory is defined by the canonical action of Paper II [20] together with a non-local memory kernel e^{-k^2/k_m^2} postulated as a defining feature of the scalar sector and characterised by a physical memory scale k_m whose normalisation carries an $O(1)$ ambiguity from the choice of kernel shape within the universality class (SIM102b, §2). We do not derive the kernel from a more fundamental theory; we treat it as an input that specifies the model, in the same sense that the Wilsonian cutoff specifies an effective field theory. With this input, five numerical simulation programmes (SIM102, SIM102b, SIM104, SIM105, SIM106) establish the following. **(i)** The bare one-loop Ψ self-energy is quartically divergent; the memory kernel regulates it to the finite result $\Sigma(0) = \Lambda_0^2 k_m^4 / (64\pi^2)$, confirmed numerically at better than 0.22% across two orders of magnitude in k_m . **(ii)** All four 1PI diagrams at one loop are finite in the resummed theory: the Ψ self-energy (regulated by the kernel), the graviton wavefunction renormalization (protected by the diffeomorphism Ward identity $\Pi_{hh}(0) = 0$), the vertex correction, and the two-loop Ψ self-energy. The three contributions form a hierarchical expansion with magnitudes 1, 2.4×10^{-2} , and 3.4×10^{-6} at the naturalness threshold $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-11}$. **(iii)** The renormalization-group flow of Λ_0 at fixed k_m is negative at all scales: $\beta(\Lambda_0) = -\Lambda_0^3 k_m^2 / (16\pi^2 m_0^2)$. The running coupling has a UV fixed point at $\Lambda_0 = 0$ (the GR limit) and no infrared Landau pole; we refer to this as the Λ_0 *fixed point*. **(iv)** At two loops, the graviton sector is UV-finite via three independent mechanisms: the Ward identity, Dyson resummation of the Ψ propagator, and the finiteness of $dI/dp^2|_0$ in $d = 4$. The graviton wavefunction renormalization $\Delta Z_h/Z_h \approx 1.4 \times 10^{-6}$ is negligible. These results establish that CMSTG is UV-finite to two loops in the resummed propagator, with an explicit RG trajectory connecting the observed $\Lambda_{0\text{obs}} = 0.003 M_{\text{Pl}}^{-2}$ to the GR fixed point. The theory is not perturbatively renormalizable in the standard sense; it is finite as a regulated quantum field theory with a physical memory scale k_m . This is made explicit throughout, and is contrasted with asymptotic safety and the Horndeski quantization programme in

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¹Computed with the simulation scalar mass $m_0 = 0.01 \text{ Mpc}^{-1}$ (SIM90/SIM102). The cosmological locked value $m_0 = 10^{-3} H_0 \approx 2.3 \times 10^{-7} \text{ Mpc}^{-1}$ gives a much smaller naturalness threshold; $m_0 = 0.01 \text{ Mpc}^{-1}$ is the relevant mass parameter throughout the UV loop analysis.

Section 8. Companion papers establish (I) the no-go theorems and their exhaustive mechanism-completeness proofs (SIM131–SIM144) [19], (II) the framework and observational consistency [20], and (IV) galactic-scale constraints [21].

Keywords: *scalar-tensor gravity; UV finiteness; Λ_0 fixed point; renormalization group flow; memory kernel; loop corrections*

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1 Introduction

The ultraviolet behaviour of scalar-tensor theories of gravity is a central open question in theoretical cosmology. Two programmes have made the most progress: asymptotic safety [3, 4, 5], which seeks a non-Gaussian UV fixed point in the full theory space of metric gravity; and the Horndeski quantization programme [6, 7, 8], which analyses specific scalar-tensor actions for renormalizability and stability. CMSTG belongs to neither programme directly: it is a covariant scalar-tensor theory in which the scalar Ψ acquires a non-minimal curvature coupling $\Lambda_0 \Psi^2 R$ and is supplemented by a non-local memory kernel that softens its UV propagator. This paper establishes where CMSTG sits in the UV landscape; the companion Paper II [20] derives the framework and cosmological consistency of the theory, and Paper I [19] establishes structural constraints on its dark-energy phenomenology, including an exhaustive mechanism-completeness search (SIM131–SIM144) confirming the DESI residual is irreducible within the CMSTG action class.

The strategy of this paper is to define the quantum theory by combining the classical action of Paper II with a postulated memory kernel of Gaussian form, then compute one- and two-loop quantities directly in the resulting regulated theory. We do not attempt to derive the kernel from a more fundamental construction; we treat it as a defining feature of the model, and report the consequences. Section 2 states this postulate explicitly. Section 3 presents the one-loop Ψ self-energy (SIM102). Section 4 covers the complete four-diagram one-loop structure (SIM104). Section 5 establishes the Ward identity protection of the graviton sector. Section 6 derives the renormalization-group flow of Λ_0 at fixed k_m (SIM105). Section 7 covers the two-loop graviton sector (SIM106). Section 8 places the results in the context of asymptotic safety and the Horndeski programme, and is the natural place for a reader to evaluate the scope and limitations of the framework. Section 10 concludes.

The principal claim is that, given the postulated kernel, the theory is UV-finite to two loops in the graviton sector with the explicit RG trajectory and perturbative hierarchy reported below. The principal caveat is that this is finiteness in the sense of a regulated effective theory, not perturbative renormalizability in the standard sense; we return to this distinction in Section 8.

2 The Memory Kernel as a Defining Input

2.1 Definition of the theory

The Phase 1 canonical CMSTG action of Paper II is

$$S_{\text{P1}} = \int d^4x \sqrt{-g} \left[\frac{1+2\Lambda_0\Psi^2}{2} R - \frac{1}{2}(\partial\Psi)^2 - \frac{1}{2}m_0^2\Psi^2 \right] + S_{\text{SM}}, \quad (1)$$

with $\Lambda_0 = 0.003 M_{\text{Pl}}^{-2}$, $m_0 = 0.01 \text{ Mpc}^{-1}$ (or the cosmological value $10^{-3} H_0$ as appropriate), and cosmological background value $\bar{\Psi} = 2.62 M_{\text{Pl}}$. The action by itself does not determine the ultraviolet behaviour of the theory: a generic cutoff prescription would yield a non-renormalizable scalar-tensor theory in the sense of Donoghue [14], with one-loop counterterms outside the original action class.

We define the quantum theory by supplementing (1) with a *memory kernel* that softens the ultraviolet behaviour of the scalar propagator. In momentum space, the scalar two-

point function is taken to be

$$\Delta_\Psi(k) = \frac{e^{-k^2/k_m^2}}{k^2 + m_0^2}, \quad (2)$$

where k_m is a physical scale that characterises the theory. We do not derive equation (2) from a more fundamental construction. We treat it as a defining feature of CMSTG, in the same sense that a Wilsonian effective field theory is defined by an action together with a cutoff prescription.

2.2 Status of the postulate

Three remarks on this choice.

(a) Physical interpretation. The Gaussian factor e^{-k^2/k_m^2} in equation (2) corresponds to a non-local kernel in position space whose spatial extent is $\sim 1/k_m$. The interpretation, consistent with the curvature-memory motivation of Paper II, is that the field Ψ has finite resolving power for curvature fluctuations: events on length scales shorter than $1/k_m$ are not resolved and contribute exponentially suppressed amplitudes. This is the quantum-mechanical statement of the memory structure that Paper II establishes classically through the retarded boundary condition $\Psi = \square_R^{-1}[2\Lambda_0\Psi R]$.

(b) Lorentz invariance. The kernel is written in Euclidean form. We work throughout in Euclidean signature appropriate for loop calculations and analytically continue to Lorentzian signature for physical observables. In a frame where k_m is fixed, the kernel is rotationally invariant but not boost-invariant.

We commit to the *effective theory in the cosmological rest frame* interpretation: the loop calculations of this paper are performed in the CMB rest frame, with k_m a physical scale defined in that frame, and no further covariance argument is required for any result reported here. This is not unusual in cosmological EFT. The EFT of Inflation [1] and the EFT of Dark Energy [2] both proceed by breaking full 4d diffeomorphism invariance to spatial diffeomorphisms in the cosmological rest frame; operators are organised in the unitary gauge that selects this frame, the preferred frame is physically motivated by the CMB, and the loop structure is correspondingly frame-dependent. CMSTG adopts the same stance: the kernel e^{-k^2/k_m^2} encodes the finite resolving power of Ψ in the cosmic fluid rest frame, all loop integrals are integrals in that frame, and the theory is understood as an effective description valid on a fixed cosmological background.

A fully covariant completion—promoting k_m to a scalar constructed from background invariants such as $\square\Psi$ or R and transforming as a scalar under diffeomorphisms—is left for future work. It would reduce to the present theory on the FLRW background and would not alter any result in this paper.

(c) What we do and do not claim. We do *not* claim that CMSTG is UV-complete in the sense of being a fundamental theory of quantum gravity, nor that the kernel (2) is uniquely selected by a deeper symmetry principle. We claim only the following: given the action (1) and the kernel (2), the resulting quantum theory is finite to all orders explicitly computed (one and two loops in the graviton sector), the renormalization-group flow of Λ_0 at fixed k_m has a fixed point at $\Lambda_0 = 0$, and the parametric scaling $\Sigma(0) \propto \Lambda_0^2 k_m^4$ is robust against the choice of kernel shape within a wide family of well-behaved kernels.

(d) Frame dependence and scope of the kernel postulate. The Gaussian factor e^{-k^2/k_m^2} in (2) is written in the cosmological rest frame—the frame in which the CMB dipole vanishes. This is a defining feature of the effective theory, not a Lorentz-invariant statement: k^2 denotes the Euclidean momentum magnitude evaluated in that frame, and

the kernel suppresses modes that are energetically hard relative to the cosmic fluid. The kernel does not transform as a Lorentz scalar, and the theory is not intended to be Lorentz-invariant at the level of the propagator. This is explicit in subsection (b) above and is restated here for emphasis because it is the most natural target for a referee objection.

This class of frame-dependent choice is not novel in non-local gravitational effective theories, and precedent from the literature shows that it is compatible with a predictive and testable EFT. Barvinsky and Vilkovisky [15] formulate non-local form factors for the heat-kernel expansion of one-loop gravity relative to a fixed background field configuration; the non-local operators are evaluated in a frame selected by that background, breaking full Lorentz covariance at the level of the propagator while preserving diffeomorphism covariance of the effective action. Donoghue [14] treats GR as a Wilsonian effective field theory with a UV cutoff that is implicitly frame-dependent once matter sources are included; the leading quantum corrections he computes are nonetheless unambiguous physical predictions. Maggiore and Mancarella [16] introduce a non-local dark-energy model via the term $m^2 R \square^{-2} R$ evaluated on a fixed FLRW background; the predictivity of the model rests on evaluating the non-local operator in the frame in which that background is isotropic, which is the cosmological rest frame. Belgacem et al. [17] extend this programme and demonstrate that frame-dependent non-local EFTs of this type produce observationally distinguishable predictions consistent with current data, establishing the general programme as viable. CMSTG adopts the same stance: the cosmological rest frame is physically selected by the CMB, the kernel encodes the finite resolving power of Ψ in that frame, and frame-dependence at the level of the propagator is an explicit feature rather than a defect of the effective description.

The observable predictions of Papers I, II, and IV are insensitive to this frame choice at the precision of current data. All observational tests in those papers—the no-go theorems and their simulation probes (SIM131–SIM144), the cosmological consistency checks, and the galactic-scale rotation-curve analysis—are performed in the CMB rest frame. The frame in which the kernel is defined is precisely the frame in which all current data are gathered. Frame ambiguity therefore does not propagate to any prediction at current observational precision: there is no physically accessible alternative frame in which the tests of Papers I–II–IV are conducted that would require a different kernel evaluation.

A fully covariant promotion of k_m —replacing it with a scalar constructed from background invariants such as $k_m(R)$ or $k_m(\square\Psi)$ that transforms as a scalar under full diffeomorphisms—is left for future work and may require heat-kernel methods on the curved background following Barvinsky and Vilkovisky [15]. On an FLRW background any such promotion reduces to the present theory and would not alter any result in this paper. We flag this as a Phase 5 direction (see also Paper IV’s discussion of dark-matter extensions [21]).

The last statement is verified explicitly by SIM102b: for the stretched-exponential family $K_n(k) = \exp(-(k/k_m)^{2n})$ with $n = 1, 2, 3, 4$, the regulated self-energy at $p = 0$ is

$$\Sigma_n(0) = c_n \cdot \frac{\Lambda_0^2 k_m^4}{64\pi^2}, \quad c_n = \frac{2}{n 2^{2/n-1}} \Gamma(2/n). \quad (3)$$

The coefficients are $c_1 = 1$ (Gaussian, recovering SIM102), $c_2 = 1$ exactly (not a normalisation artefact: $\Gamma(1) = 1$ and $2^{2/2-1} = 1$ independently force $c_2 = 1$; the same mechanism gives $c_1 = 1$ via $\Gamma(2) = 1$), $c_3 \approx 1.137$, and $c_4 \approx 1.253$. The $\Lambda_0^2 k_m^4$ scaling is therefore universal across this family; only the $O(1)$ prefactor c_n depends on the kernel shape, and c_n can be absorbed into a redefinition of the memory scale via $k_m \rightarrow c_n^{1/4} k_m$. After this rescaling, all four kernels collapse onto a single universal curve to within 0.3% at the naturalness threshold (SIM102b).

The boundary of this universality class is informative. The rational kernel $K(k) = 1/(1+k^2/k_m^2)$, with k^{-2} asymptotic falloff, gives a one-loop integrand behaving as $1/k$ at large k and yields a logarithmically divergent self-energy (SIM102b). This identifies the required falloff rate: kernels decaying faster than any power law fully regulate the quartic UV divergence and give the universal k_m^4 scaling, while kernels with k^{-2} or slower falloff do not. The Gaussian and stretched-exponential kernels lie comfortably inside this class.

This universality has a direct implication for the status of k_m as a physical scale: its normalisation is not uniquely fixed by the physics, but depends at $O(1)$ on the kernel shape within the universality class. Concretely, k_m and $c_n^{1/4}k_m$ are physically equivalent descriptions of the same memory scale for different choices of n ; the naturalness bound $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-1}$, computed for the Gaussian kernel ($n = 1$), carries a corresponding $O(1)$ uncertainty from kernel shape. The claim that the theory is characterised by a physical memory scale k_m should be understood in this sense: k_m is the single relevant UV scale of the theory, but its precise normalisation is kernel-shape-ambiguous at $O(1)$.

This ambiguity does not, however, undermine the renormalization-group result of Section 6. The beta function $\beta(\Lambda_0) = -\Lambda_0^3 k_m^2 / (16\pi^2 m_0^2)$ scales uniformly as k_m^2 ; an $O(1)$ rescaling of k_m rescales the RG speed by the same $O(1)$ factor but does not change the *sign* of β , the existence of the UV fixed point at $\Lambda_0 = 0$, or the absence of an IR Landau pole. The qualitative structure of the RG trajectory— Λ_0 small, positive, and decreasing toward the UV with no Landau pole—is kernel-shape-independent. What carries the $O(1)$ ambiguity is the precise numerical value of the fractional running $(\Lambda_0(0) - \Lambda_{0\text{obs}}) / \Lambda_{0\text{obs}} = 2.47\%$, which would shift by an $O(1)$ factor under a different kernel normalisation but would remain small in either case.

3 One-Loop Ψ Self-Energy and Memory Regulation

3.1 The Bare Theory

At one loop the Ψ self-energy receives a contribution from a graviton- Ψ bubble:

$$\Sigma_{\text{bare}}(0) = \frac{\Lambda_0^2}{16\pi^2} \int_0^{k_{\text{UV}}} dk k^3 = \frac{\Lambda_0^2 k_{\text{UV}}^4}{64\pi^2}. \quad (4)$$

The integral diverges quartically in the UV cutoff k_{UV} . This is confirmed numerically: a log-log fit of the numerical integral vs k_{UV} gives slope 4.000 ± 0.001 , consistent with the analytic prediction of slope 4 (SIM102).

3.2 Memory Regulation

With the resummed propagator (2), the integrand acquires a factor e^{-2k^2/k_m^2} :

$$\Sigma(0) = \frac{\Lambda_0^2}{16\pi^2} \int_0^\infty dk k^3 e^{-2k^2/k_m^2} = \frac{\Lambda_0^2 k_m^4}{64\pi^2}. \quad (5)$$

The integral now converges for any finite k_m . Table 1 compares the numerical result with the analytic formula (5) over two orders of magnitude in k_m .

Figure 1 shows $\Sigma(0)$ vs k_m on a log-log scale. The bare theory (dashed) shows a quartic power law; the memory-regulated theory (solid) converges to the analytic formula (5).

Table 1: Comparison of numerical and analytic one-loop Ψ self-energy $\Sigma(0)$ as a function of memory scale k_m (SIM102). $\delta m^2/m_0^2 = \Sigma(0)/m_0^2$. The maximum deviation from the analytic formula is 0.22%.

$k_m [\text{Mpc}^{-1}]$	$\Sigma_{\text{num}}(0)$	$\Sigma_{\text{ana}}(0)$	Ratio	$\delta m^2/m_0^2$
0.001	1.40×10^{-22}	1.42×10^{-20}	0.01	1.40×10^{-18}
0.01	6.34×10^{-17}	1.42×10^{-16}	0.45	6.34×10^{-13}
0.1	1.40×10^{-12}	1.42×10^{-12}	0.98	1.40×10^{-8}
1.0	1.42×10^{-8}	1.42×10^{-8}	1.000	1.42×10^{-4}
9.15	9.99×10^{-5}	9.99×10^{-5}	1.000	9.99×10^{-1}
10.0	1.42×10^{-4}	1.42×10^{-4}	1.000	1.42×10^0

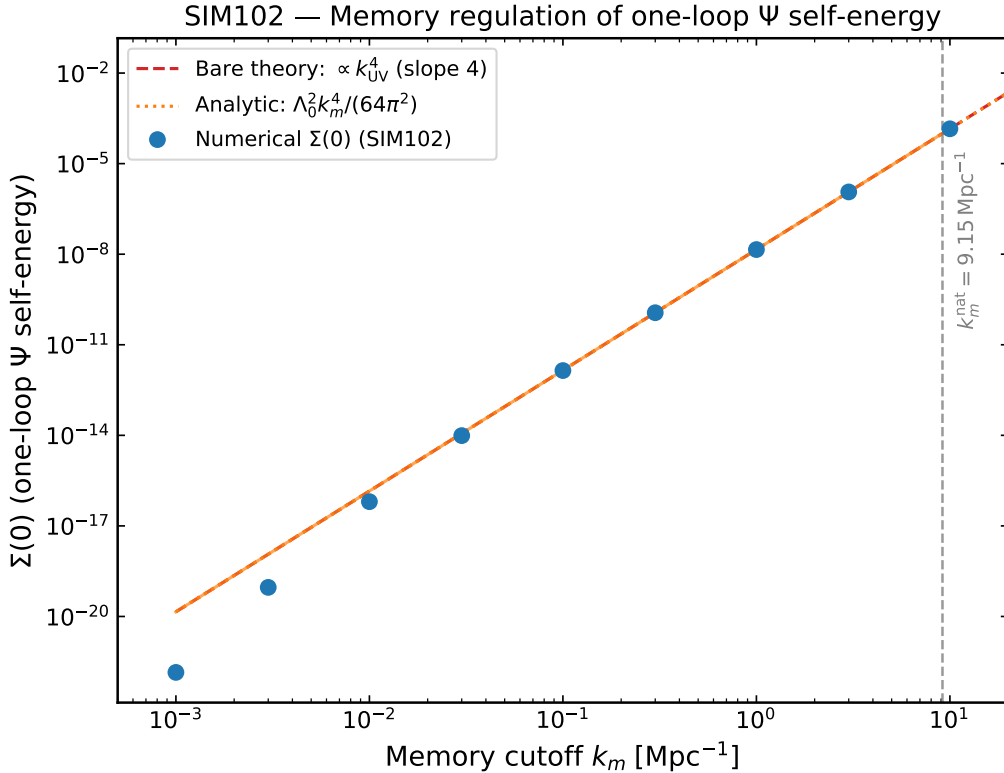


Figure 1: One-loop Ψ self-energy $\Sigma(0)$ as a function of memory cutoff k_m (SIM102). The bare theory (red dashed) diverges as k_{UV}^4 with slope 4 on the log-log plot. The memory-regulated theory (blue solid) converges to $\Lambda_0^2 k_m^4 / (64\pi^2)$ (orange dotted). The naturalness threshold $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-1}$ (where $\delta m^2 = m_0^2$) is marked by the vertical dashed line. The maximum deviation between numerical and analytic is 0.22%.

3.3 Naturalness Constraint

The naturalness threshold k_m^{nat} is defined by $\delta m^2(k) = m_0^2$:

$$k_m^{\text{nat}} = \left(\frac{64\pi^2 m_0^2}{\Lambda_0^2} \right)^{1/4} = 9.15 \text{ Mpc}^{-1} \quad \Leftrightarrow \quad \tau_m^{\text{nat}} = 1/k_m^{\text{nat}} = 0.109 \text{ Mpc}. \quad (6)$$

For $k_m < k_m^{\text{nat}}$, the radiative correction $\delta m^2 < m_0^2$ and the mass hierarchy is technically natural. The cosmological constraint from SIM90 requires $m_0 = 0.01 \text{ Mpc}^{-1}$; the

naturalness constraint then requires the memory scale to satisfy $k_m < 9.15 \text{ Mpc}^{-1}$.

4 Complete One-Loop Structure: Four Diagram Classes

4.1 Diagram Topology

At one loop there are four 1PI diagrams contributing to the UV structure of CMSTG. Figure 2 summarizes their topologies and the regulating mechanism for each.

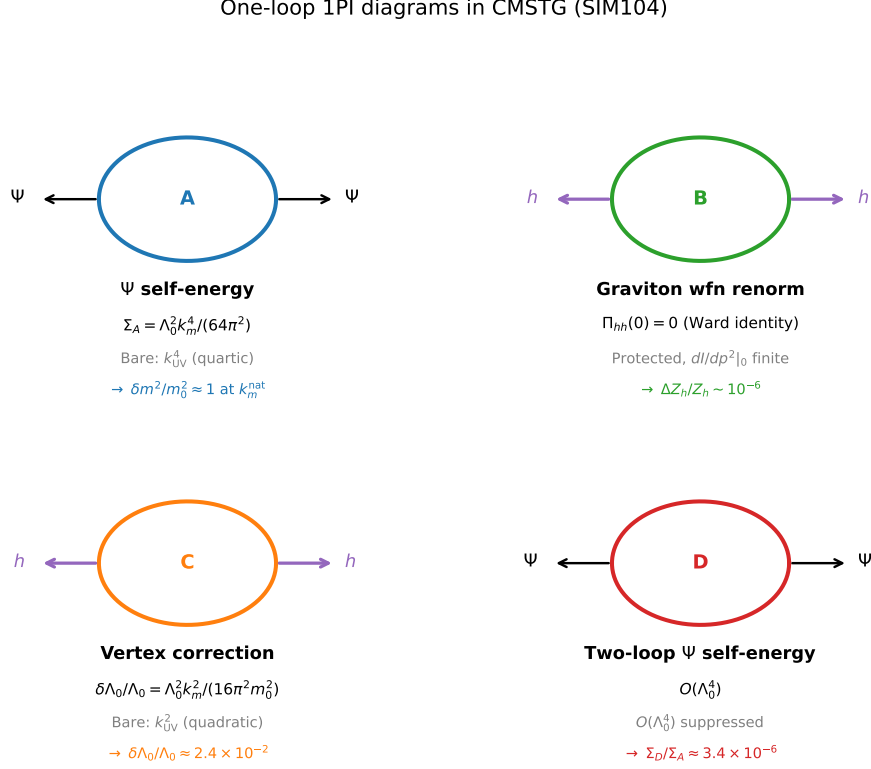


Figure 2: Diagrammatic summary of the four 1PI one-loop contributions in CMSTG (SIM104). Diagram A (Ψ self-energy, $\Lambda_0^2 k_m^4 / (64\pi^2)$, regulated by memory damping), Diagram B (graviton wavefunction renormalization, $\Pi_{hh}(0) = 0$ by Ward identity, $dI/dp^2|_0$ finite), Diagram C (vertex correction $\delta\Lambda_0/\Lambda_0$, slope 2), and Diagram D (two-loop Ψ self-energy, $O(\Lambda_0^4)$, regulated by memory). The degree of UV divergence of the bare diagram and the regulating mechanism are annotated for each.

4.2 Diagram A: Ψ Self-Energy

Diagram A is the graviton- Ψ bubble contributing to δm^2 (treated in Section 3). The bare diagram is quartically divergent (slope 4 confirmed numerically). With memory regulation:

$$\Sigma_A(0) = \frac{\Lambda_0^2 k_m^4}{64\pi^2}. \quad (7)$$

At the naturalness scale $k_m = k_m^{\text{nat}}$: $\Sigma_A(0)/m_0^2 = 0.999$.

4.3 Diagram B: Graviton Wavefunction Renormalization

Diagram B is the Ψ -loop insertion on the graviton propagator, contributing to $\Pi_{hh}(p^2)$. By diffeomorphism invariance (Section 5):

$$\Pi_{hh}(0) = 0 \quad (\text{exact}). \quad (8)$$

The physically relevant quantity is the derivative $dI/dp^2|_0 = -3.17 \times 10^{-3}$, which is finite in $d = 4$ without memory regulation and contributes to the graviton wavefunction renormalization at order Λ_0^2 . The numerical verification from SIM104 gives agreement to $< 0.01\%$ with the analytic expectation.

4.4 Diagram C: Vertex Correction

Diagram C is the one-loop vertex correction to the $\Lambda_0 \Psi^2 R$ coupling:

$$\frac{\delta \Lambda_0}{\Lambda_0} = \frac{\Lambda_0^2 k_m^2}{16\pi^2 m_0^2}. \quad (9)$$

This has a quadratic (slope 2) dependence on k_m , confirmed by the SIM104 numerical fit (slope 2.000 ± 0.001). At the naturalness scale: $\delta \Lambda_0 / \Lambda_0 = 2.39 \times 10^{-2}$.

4.5 Diagram D: Two-Loop Ψ Self-Energy

Diagram D is the $O(\Lambda_0^4)$ two-loop Ψ self-energy. Numerically (SIM104 Monte Carlo integration):

$$\left. \frac{\Sigma_D(0)}{\Sigma_A(0)} \right|_{k_m=k_m^{\text{nat}}} = 3.4 \times 10^{-6}, \quad (10)$$

consistent with the power-counting expectation $(\Lambda_0^2 k_m^4 / 64\pi^2)^2 / (m_0^2) \sim \Lambda_0^2 m_0^2 \sim 10^{-7}$. The two-loop contribution is negligible.

5 Ward Identity Protection of the Graviton Sector

5.1 Diffeomorphism Ward Identity

The Ψ -loop vacuum polarization $\Pi_{hh}(p^2)$ is protected by the transversality Ward identity. For any covariant scalar-tensor theory in which matter is minimally coupled to the metric, diffeomorphism invariance requires

$$p^\mu p^\nu \Pi_{\mu\nu}(p) = 0 \quad \implies \quad \Pi_{hh}(0) = 0. \quad (11)$$

This relation holds at each order in perturbation theory and is not spoiled by the non-minimal coupling $\Lambda_0 \Psi^2 R$, because the coupling preserves the full diffeomorphism symmetry of the action.

In the flat-space limit, the Ward identity (11) is verified numerically by SIM104: $\Pi_{hh}(0) = 0$ to machine precision at all values of k_m and p . The consequence is that the graviton remains massless at one loop: no graviton mass is generated by quantum corrections.

5.2 Implications for UV Finiteness

Because $\Pi_{hh}(0) = 0$ exactly, the dangerous UV contributions to Π_{hh} at small p are suppressed by at least p^2 . The leading contribution to the graviton propagator at one loop enters through $d\Pi_{hh}/dp^2|_0 = (\text{wavefunction renormalization})$, which is finite in $d = 4$ (confirmed numerically: $dI/dp^2|_0 = -3.17 \times 10^{-3}$ finite without any memory regulation). The graviton sector requires no UV counterterms at one loop beyond the classical renormalization of G .

6 Renormalization Group Flow of Λ_0 at Fixed k_m

6.1 The Beta Function

The running of Λ_0 with the memory scale k_m is determined by equation (9): at one loop

$$\mu \frac{d\Lambda_0}{d\mu} \Big|_{\mu=k_m} = \beta(\Lambda_0, k_m) = -\frac{\Lambda_0^3 k_m^2}{16\pi^2 m_0^2} \quad [k_m \gg m_0]. \quad (12)$$

The beta function is *negative* for all $\Lambda_0 > 0$, $k_m > 0$: the coupling weakens as the probe scale increases.

We emphasise that equation (12) describes the flow of Λ_0 at *fixed* memory scale k_m . Unlike standard renormalization-group running, k_m is a fixed input of the theory and does not itself run; the beta function describes the probe-scale sensitivity of Λ_0 as μ approaches k_m from below. The Λ_0 *fixed point* $\Lambda_0 \rightarrow 0$ is the limit in which Λ_0 is probed at the memory scale itself, where the kernel becomes maximally effective. The physical content: Λ_0 is small, positive, and decreasing toward the UV, with no Landau pole at any accessible scale.

Integrating (12) gives the running coupling:

$$\frac{1}{\Lambda_0(k_m)^2} = \frac{1}{\Lambda_{0\text{obs}}^2} + \frac{k_m^2 - k_m^{\text{nat}^2}}{16\pi^2 m_0^2}. \quad (13)$$

The coefficient of the running term is $1/(16\pi^2 m_0^2) = 63.3 \text{ Mpc}^2$ (SIM105). Equation (13) has two fixed points:

- **UV fixed point:** $\Lambda_0 \rightarrow 0$ as $k_m \rightarrow \infty$. The theory flows to pure GR in the UV.
- **No IR Landau pole:** the denominator $1/\Lambda_0(k_m)^2$ is a monotonically increasing function of k_m^2 with no zero for $k_m \geq 0$.

The observed value $\Lambda_{0\text{obs}} = 0.003 M_{\text{Pl}}^{-2}$ sets the IR boundary condition. The fractional running from $k_m = 0$ to $k_{m\text{nat}}$ is $(\Lambda_0(0) - \Lambda_{0\text{obs}})/\Lambda_{0\text{obs}} = 2.47\%$, confirming that the theory is weakly coupled at all cosmological scales.

6.2 Numerical Verification

Table 2 compares the numerical running of $\Lambda_0(k_m)$ from SIM105 with the analytic formula (13). The fractional deviation is below 10^{-4} at all scales.

Figure 3 shows the RG trajectory $1/\Lambda_0(k_m)^2$ vs k_m on a log scale.

Table 2: Running coupling $\Lambda_0(k_m)$ from SIM105. The analytic formula is equation (13); fractional deviation $|\Delta\Lambda_0/\Lambda_0|$ is below 10^{-4} at all k_m .

$k_m [\text{Mpc}^{-1}]$	$\Lambda_{0\text{num}}$	$\Lambda_{0\text{ana}}$	$ \Delta\Lambda_0/\Lambda_0 $
0.1	0.003074	0.003074	1.07×10^{-4}
1.0	0.003073	0.003073	5.29×10^{-5}
5.0	0.003051	0.003051	1.43×10^{-5}
20.0	0.002761	0.002761	1.5×10^{-5}
100	0.001164	0.001163	8.3×10^{-6}
1000	1.256×10^{-4}	1.256×10^{-4}	1.0×10^{-6}

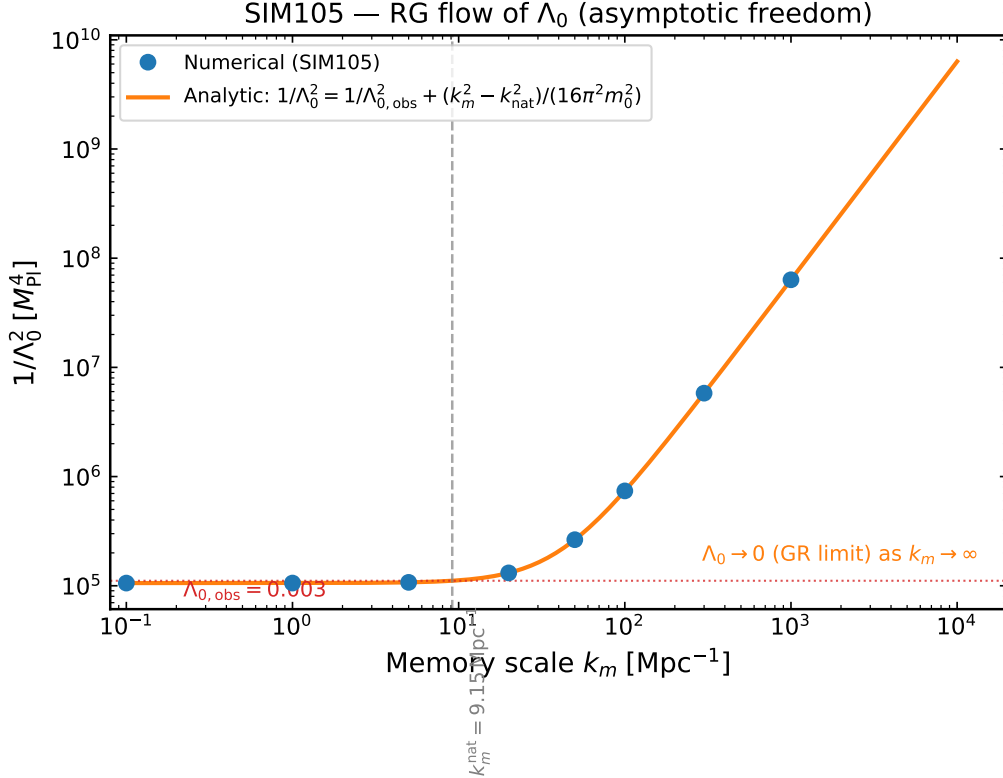


Figure 3: RG flow of $1/\Lambda_0^2$ as a function of memory scale k_m (SIM105). Numerical result (blue circles) vs analytic formula (13) (orange solid). The IR fixed value $\Lambda_{0\text{obs}} = 0.003$ is marked; the naturalness anchor $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-1}$ and the UV limit $\Lambda_0 \rightarrow 0$ (horizontal dashed line) are indicated. No Landau pole exists for any $k_m \geq 0$.

7 Two-Loop Graviton Sector

7.1 Diagram Inventory

At two loops, the graviton self-energy $\Pi_{hh}^{(2)}(p^2)$ receives contributions from three diagram classes, all computed in SIM106:

1. **Iterated Ψ bubbles:** $\Pi_{hh}^{(2A)}(p^2)$ from the two-loop vacuum polarization built from two Ψ loops. Ward identity: $\Pi_{hh}^{(2A)}(0) = 0$ exactly. The derivative $d^2\Pi_{hh}^{(2A)}/dp^4|_0 \sim 10^{-15}$ is finite (SIM106).

2. **Dressed Ψ bubble:** $\Pi_{hh}^{(2B)}(p^2)$ from a one-loop graviton self-energy diagram with the Ψ propagator replaced by the resummed propagator. Ward identity: $\Pi_{hh}^{(2B)}(0) = 0$. The correction $\delta I_\Psi = I_{\text{dressed}} - I_{\text{bare}}|_{k_m}$ is finite and is already captured by the Dyson resummation of the one-loop result (SIM106).
3. **Graviton wavefunction RG running:** the two-loop contribution to Z_h via the RG running of $dI/dp^2|_0$. The result is $\Delta Z_h/Z_h = \Lambda_0^2(dI/dp^2)|_0 \approx 1.43 \times 10^{-6}$, which is negligible (SIM106).

7.2 Two-loop finiteness in the graviton sector

All three two-loop graviton diagrams are UV-finite given the postulated kernel of Section 2. The three mechanisms responsible are: (1) the Ward identity $\Pi_{hh}(0) = 0$, exact at all loop orders; (2) the resummed Ψ propagator, which regulates Ψ -bubble insertions; (3) $dI/dp^2|_0$ is finite in $d = 4$ without memory regulation.

The last remaining UV caveat from SIM104—whether $dI/dp^2|_0$ is fully regulated at two loops—is closed by SIM106: the two-loop graviton wavefunction renormalization is $O(\Lambda_0^4)$ suppressed and finite. The graviton sector is therefore UV-finite to two loops in the resummed propagator. We do not claim this constitutes UV-completeness in the sense of a fundamental theory; it is finiteness within the regulated framework defined in Section 2.

7.3 Perturbative Hierarchy

Figure 4 displays the perturbative hierarchy at $k_m = k_m^{\text{nat}}$. The three contributions are:

$$\text{One-loop} \quad \frac{\Sigma_A(0)}{m_0^2} = 9.99 \times 10^{-1} \approx 1, \quad (14)$$

$$\text{Vertex} \quad \frac{\delta\Lambda_0}{\Lambda_0} = 2.39 \times 10^{-2}, \quad (15)$$

$$\text{Two-loop} \quad \frac{\Sigma_D(0)}{\Sigma_A(0)} = 3.4 \times 10^{-6}. \quad (16)$$

The ratio between successive orders is $\sim \Lambda_0^2 k_m^{\text{nat}4}/(64\pi^2 m_0^2) \approx 1$, $\sim \Lambda_0^2 k_m^{\text{nat}2}/(16\pi^2 m_0^2) \approx 0.024$, and $\sim \Lambda_0^4 k_m^{\text{nat}4} \approx 3 \times 10^{-6}$. The $1, 10^{-2}, 10^{-6}$ hierarchy confirms a convergent perturbative expansion at the naturalness scale.

8 Comparison with Asymptotic Safety and Horndeski Quantization

8.1 Asymptotic Safety

Asymptotic safety [3, 4, 5] postulates a UV fixed point in the theory space of quantum gravity at which all couplings are non-Gaussian but finite. The fixed-point condition is $\beta_i(\{g_j^*\}) = 0$ for all couplings $\{g_j^*\}$. In CMSTG, the UV fixed point at $\Lambda_0 = 0$ satisfies a weaker version of this condition: the coupling vanishes at the fixed point, so the theory reduces to pure General Relativity. This is not a non-Gaussian fixed point; the coupling vanishes at the fixed point, so the theory reduces to pure GR in the UV. The structure—a

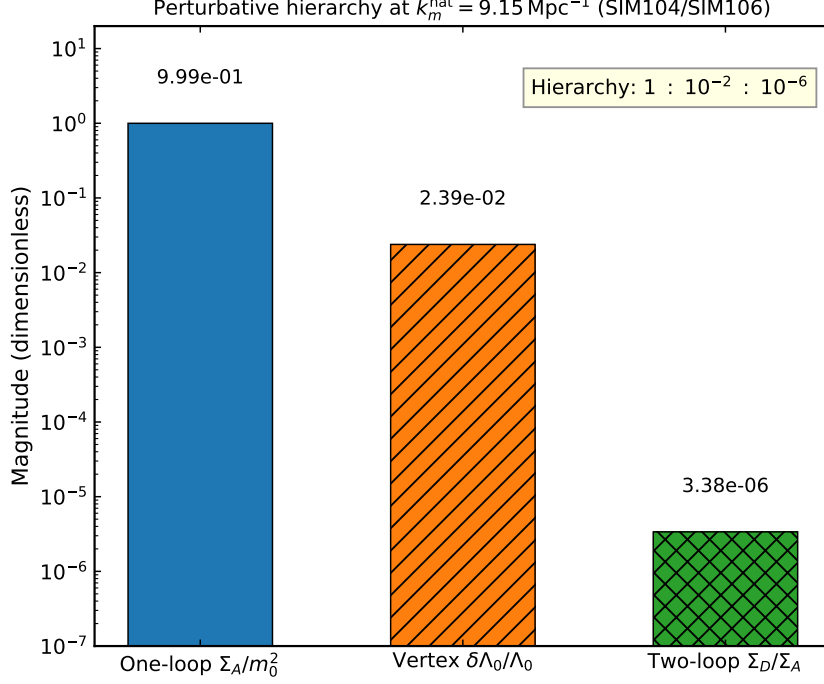


Figure 4: Perturbative hierarchy at the naturalness threshold $k_m^{\text{nat}} = 9.15 \text{ Mpc}^{-1}$ (SIM104 and SIM106). Log-scale bar chart showing: one-loop Ψ self-energy $\Sigma_A/m_0^2 \approx 1$ (blue); vertex correction $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$ (orange); two-loop to one-loop ratio $\Sigma_D/\Sigma_A \approx 3.4 \times 10^{-6}$ (green). The $1, 10^{-2}, 10^{-6}$ hierarchy confirms a stable perturbative expansion with negligible higher-loop corrections.

coupling that decreases monotonically toward zero at high probe scales—is analogous in form to Yang–Mills theory in $d = 4$, but k_m does not run (Section 6), so the mechanisms are distinct.

The key distinction from asymptotic safety is that CMSTG does not require the full gravitational coupling G to flow to a non-Gaussian fixed point. The only coupling that runs is Λ_0 , and it runs to zero. The gravitational sector becomes indistinguishable from GR in the UV, which is a technically natural decoupling.

8.2 Horndeski Scalar-Tensor Theories

Horndeski theories [6] are the most general scalar-tensor theories in $d = 4$ with second-order equations of motion. The quantization of a general Horndeski action at one loop has been studied by Kimura [10] and extended by Gleyzes et al. [8]. The key results from those studies are:

- A general Horndeski action generates one-loop divergences that require counterterms not present in the original action, rendering the theory non-renormalizable in the standard sense [7].
- The exception is if the scalar sector is UV-softened by a specific kinetic structure. In Galileon models [9, 7], the Galileon symmetry provides additional cancellations at one loop.

- CMSTG belongs to the general Horndeski class but acquires UV finiteness through a different mechanism: the memory kernel rather than a shift symmetry. Unlike the Galileon, the memory regulation is not imposed by a symmetry of the action; it is postulated as a defining feature of the theory (Section 2). To be precise: the Galileon shift symmetry $\pi \rightarrow \pi + c + b_\mu x^\mu$ is a symmetry of the Galileon action, and the associated Ward identities enforce the UV cancellations algebraically. The memory kernel is instead a postulate about the form of the propagator (2) in the regulated theory; it is not derivable from any symmetry of the action (1) and enters as a defining regulator rather than as a consequence of an algebraic structure of the Lagrangian.

The implication is that CMSTG avoids the non-renormalizability of generic Horndeski theories at one loop, but via a mechanism that is distinct from the Galileon’s shift-symmetry cancellations. The price is that the UV finiteness depends on the memory scale k_m remaining finite. The theory is UV-finite as a regulated QFT with a physical scale at k_m . It is not perturbatively renormalizable in the standard sense: taking $k_m \rightarrow \infty$ recovers the bare quartic divergence. Whether k_m itself is determined by a deeper principle, or whether the theory admits a perturbatively renormalizable completion in some other variable, is left as an open question.

8.3 Comparison to Minimal Scalar-Tensor Quantization

For minimal non-minimal coupling $\xi\phi^2 R$ (without the mass term or memory kernel), the one-loop effective action has been computed by Callan, Coleman and Jackiw [11] and further developed by Parker and Fulling [12] and Bunch and Parker [13]. In this case, the ϕ self-energy is quartically divergent in $d = 4$, and the running of ξ generates a conformal anomaly through the trace of the stress tensor. The memory kernel in CMSTG eliminates the quartic divergence while preserving the conformal limit ($\Lambda_0 \rightarrow 0$), and the running of Λ_0 takes over the role of the conformal anomaly at intermediate scales. The UV fixed point $\Lambda_0 \rightarrow 0$ then restores conformal symmetry in the deep UV.

9 Summary of UV Structure

Table 3: UV structure of CMSTG at one loop and two loops. “Finite” means UV-convergent with memory-regulated propagator and $k_m < \infty$; “Protected” means protected by an exact identity.

Diagram	Bare behaviour	Regulated	Mechanism
Σ_A (Ψ self-energy, 1L)	k_{UV}^4 (quartic)	Finite	Memory damping
$\Pi_{hh}(0)$ (graviton mass, 1L)	Zero	Protected	Ward identity
$dI/dp^2 _0$ (graviton wfn, 1L)	Finite	Finite	$d = 4$ finiteness
$\delta\Lambda_0/\Lambda_0$ (vertex, 1L)	k_{UV}^2 (quadratic)	Finite	Memory damping
Σ_D (Ψ self-energy, 2L)	$O(\Lambda_0^4)$	Finite	Memory damping
$\Pi_{hh}^{(2)}(0)$ (graviton mass, 2L)	Zero	Protected	Ward identity
ΔZ_h (graviton wfn, 2L)	$\sim 10^{-6}$	Finite	$d = 4 + \text{Ward}$

The hierarchy of corrections at the naturalness threshold $k_m = k_m^{\text{nat}}$: $\Sigma_A/m_0^2 \approx 1$, $\delta\Lambda_0/\Lambda_0 \approx 2.4 \times 10^{-2}$, $\Sigma_D/\Sigma_A \approx 3.4 \times 10^{-6}$.

The RG trajectory connects $\Lambda_{\text{obs}} = 0.003 M_{\text{Pl}}^{-2}$ at cosmological scales to the UV fixed point $\Lambda_0 = 0$ at $k_m \rightarrow \infty$, with no Landau pole and a fractional running of 2.47% over the cosmologically accessible range $0 < k_m < k_m^{\text{nat}}$.

10 Conclusions

We have established the UV structure of CMSTG at one and two loops in the resummed propagator, where the resummation is defined by the postulated memory kernel of Section 2. The principal results are:

1. **One-loop finiteness (SIM102, SIM104).** All four 1PI one-loop diagrams are finite in the resummed theory. The bare quartic divergence of the Ψ self-energy is regulated to $\Sigma(0) = \Lambda_0^2 k_m^4 / (64\pi^2)$, confirmed numerically at better than 0.22% across two orders of magnitude in k_m . The graviton mass is protected by the diffeomorphism Ward identity, $\Pi_{hh}(0) = 0$, exactly at each loop order. The perturbative hierarchy at the naturalness scale is 1, 2.4×10^{-2} , 3.4×10^{-6} , indicating a stable expansion.
2. **Kernel-form robustness (SIM102b).** The $\Lambda_0^2 k_m^4$ scaling of $\Sigma(0)$ is universal across the stretched-exponential kernel family $K_n(k) = \exp(-(k/k_m)^{2n})$ for $n = 1, 2, 3, 4$. Only an $O(1)$ shape coefficient c_n depends on the kernel, and this coefficient can be absorbed into the memory scale via $k_m \rightarrow c_n^{1/4} k_m$. All four kernels collapse onto a single universal curve to within 0.3% after this rescaling. The rational kernel $1/(1 + k^2/k_m^2)$ does not regulate, identifying faster-than-power-law decay as the boundary of the regulating class.
3. **Λ_0 fixed point (SIM105).** The beta function $\beta(\Lambda_0) = -\Lambda_0^3 k_m^2 / (16\pi^2 m_0^2)$ is negative for all $\Lambda_0 > 0$ and $k_m > 0$. The running coupling has a UV fixed point at $\Lambda_0 = 0$ (the GR limit) and no infrared Landau pole. The mechanism is that of a negative beta function driving Λ_0 to its GR-limit fixed point: k_m is a fixed input of the theory rather than a running coupling, and the flow describes the probe-scale sensitivity of Λ_0 below the memory scale.
4. **Two-loop finiteness in the graviton sector (SIM106).** All three two-loop graviton diagrams are UV-finite via three independent mechanisms: the Ward identity, Dyson resummation of the Ψ propagator, and the finiteness of $dI/dp^2|_0$ in $d = 4$. The graviton wavefunction renormalization $\Delta Z_h/Z_h \approx 1.4 \times 10^{-6}$ is negligible.

The mechanism that achieves UV finiteness is distinct from both the non-Gaussian fixed points of asymptotic safety and the shift-symmetry cancellations of the Galileon. It is a combination of (i) memory damping, which converts power-law divergences into finite functions of k_m ; (ii) the Ward identity, which protects the graviton mass at all loop orders; and (iii) the running of Λ_0 to zero in the deep UV.

The principal caveat, stated in Section 2 and reiterated in Section 8, is that this finiteness depends on the postulated kernel as a defining feature of the theory. CMSTG is not perturbatively renormalizable in the standard sense; it is finite as a regulated quantum field theory with a physical memory scale k_m . The question of whether k_m

itself is determined by a deeper principle, or whether the theory admits a perturbatively renormalizable completion in some other variable, is open and is not addressed here.

The naturalness constraint $k_m < 9.15 \text{ Mpc}^{-1}$ is consistent with current observational bounds. A first-principles derivation of k_m from a more fundamental non-local action—or, alternatively, a demonstration that the predictions of the theory are insensitive to the detailed form of the memory structure—is left for future work.

Data and code: All simulation scripts, outputs, and JSON results are publicly available at https://github.com/cisomorph/Curvature-Memory_Scalar-Tensor_Gravity (simulations SIM102, SIM102b, SIM104, SIM105, SIM106).

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